PRACTICAL 10

Consider a disk queue with requests for I/O to blocks on cylinders
98, 183, 41, 122, 14, 124, 65, 67.

The SSTF (Shortest Seek Time First) scheduling algorithm is used. The disk head is initially at cylinder number 53. The cylinders are numbered from 0 to 199.

Write a program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

```
  UW PICO 5.09

#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[100], visited[100] = {0};
    int n, head, i, count = 0;
    int total_movement = 0;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    printf("Enter request sequence:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &requests[i]);
    }

    printf("Enter initial head position: ");
    scanf("%d", &head);

    while(count < n) {
        int min = 1000, index = -1;

        for(i = 0; i < n; i++) {
            if(!visited[i]) {
                int distance = abs(head - requests[i]);
                if(distance < min) {
                    min = distance;
                    index = i;
                }
            }
        }

        visited[index] = 1;
        total_movement += min;
        head = requests[index];
        count++;

        printf("Head moves to %d (Movement = %d)\n", head, min);
    }

    printf("\nTotal Head Movement = %d\n", total_movement);

    return 0;
}
```

```
(base) sakshirakhade@Sakshis-MacBook-Air ~ % nano sstf.c
(base) sakshirakhade@Sakshis-MacBook-Air ~ % gcc sstf.c -o sstf
(base) sakshirakhade@Sakshis-MacBook-Air ~ % ./sstf
Enter number of requests: 8
Enter request sequence:
98 183 41 122 14 124 65 67
Enter initial head position: 53
Head moves to 41 (Movement = 12)
Head moves to 65 (Movement = 24)
Head moves to 67 (Movement = 2)
Head moves to 98 (Movement = 31)
Head moves to 122 (Movement = 24)
Head moves to 124 (Movement = 2)
Head moves to 183 (Movement = 59)
Head moves to 14 (Movement = 169)

Total Head Movement = 323
(base) sakshirakhade@Sakshis-MacBook-Air ~ %
```